SMARTX: Porewater pH

September 2025 Samples

2025-10-22

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##Run Information

```r
cat("Run Information:  ") #lets you know what section you're in
```

```
## Run Information:
```

```r
#set the run date & user name
  sample_year <- "2025"
  sample_month <- "September"
  user <- "Victoria Ellis"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  pH_file <- "Raw Data/SMARTX_pH_(September 610).csv"

# Define the file path for  file - check year and month
  final_path <- "Processed Data/GCReW_SMARTX_Porewater_pH_202509.csv"

#record any notes about the run or anything other info here:
  run_notes <- ""

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

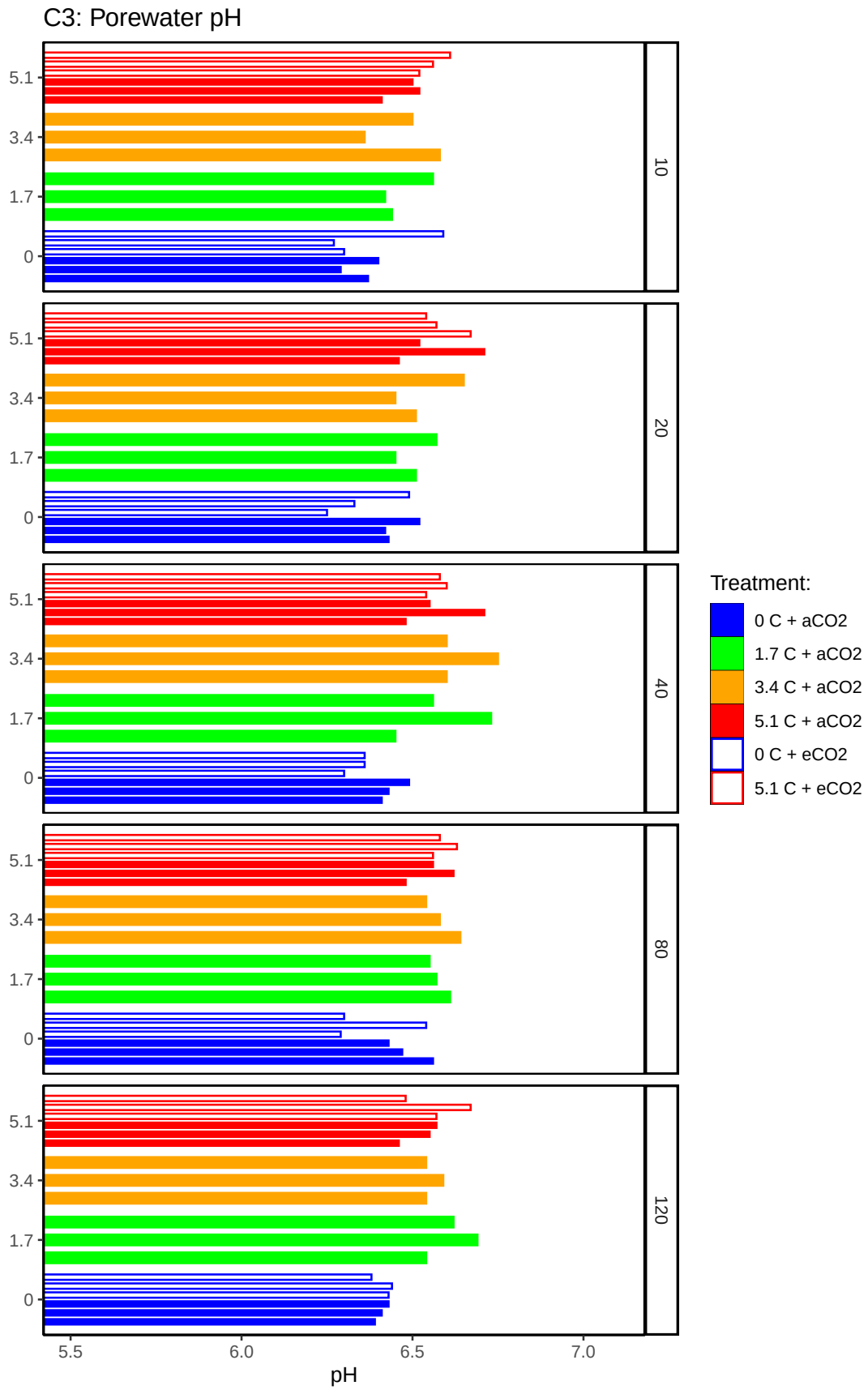
0.1 Merge samples with Metadata & check all samples are present

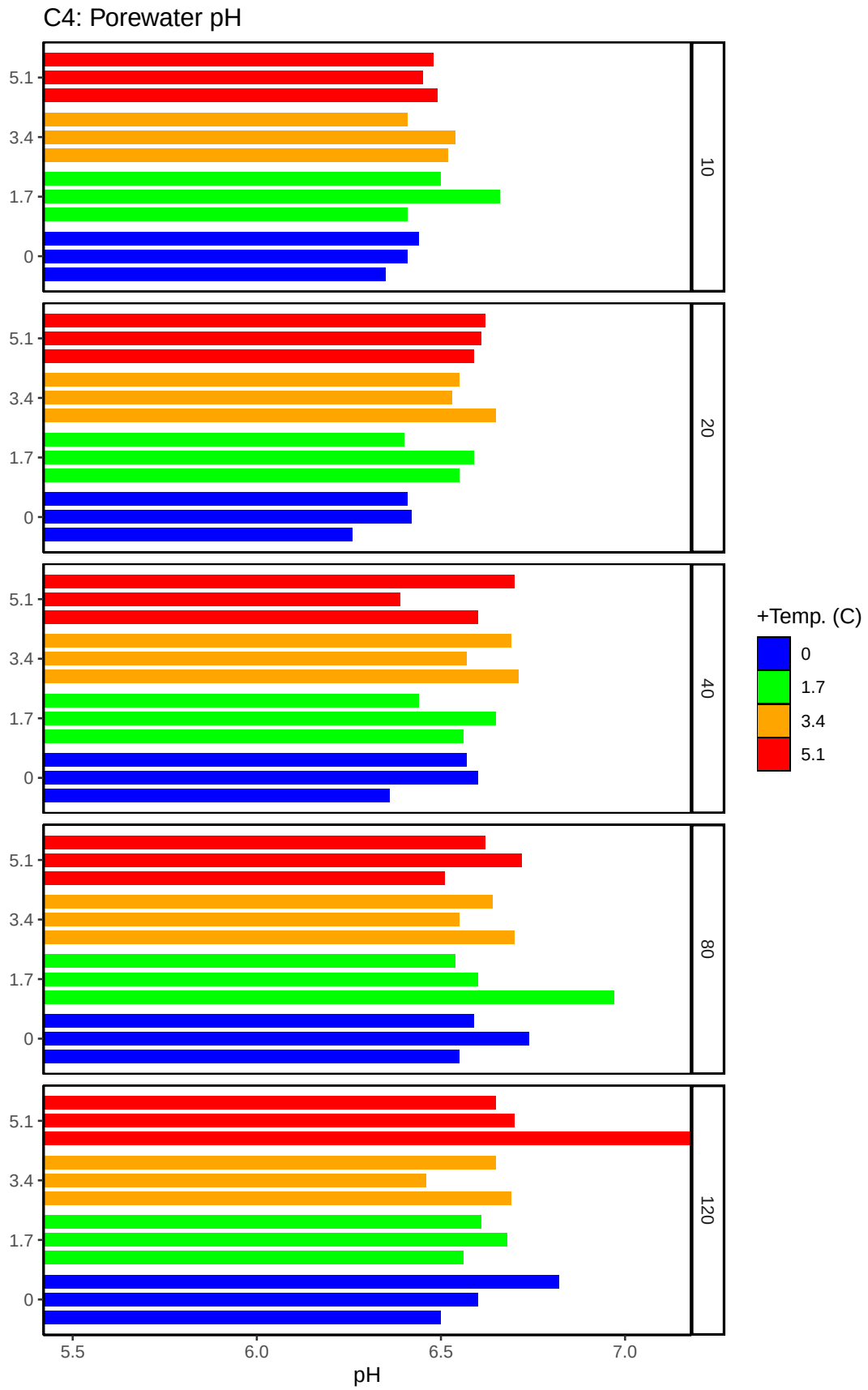
```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data.
```

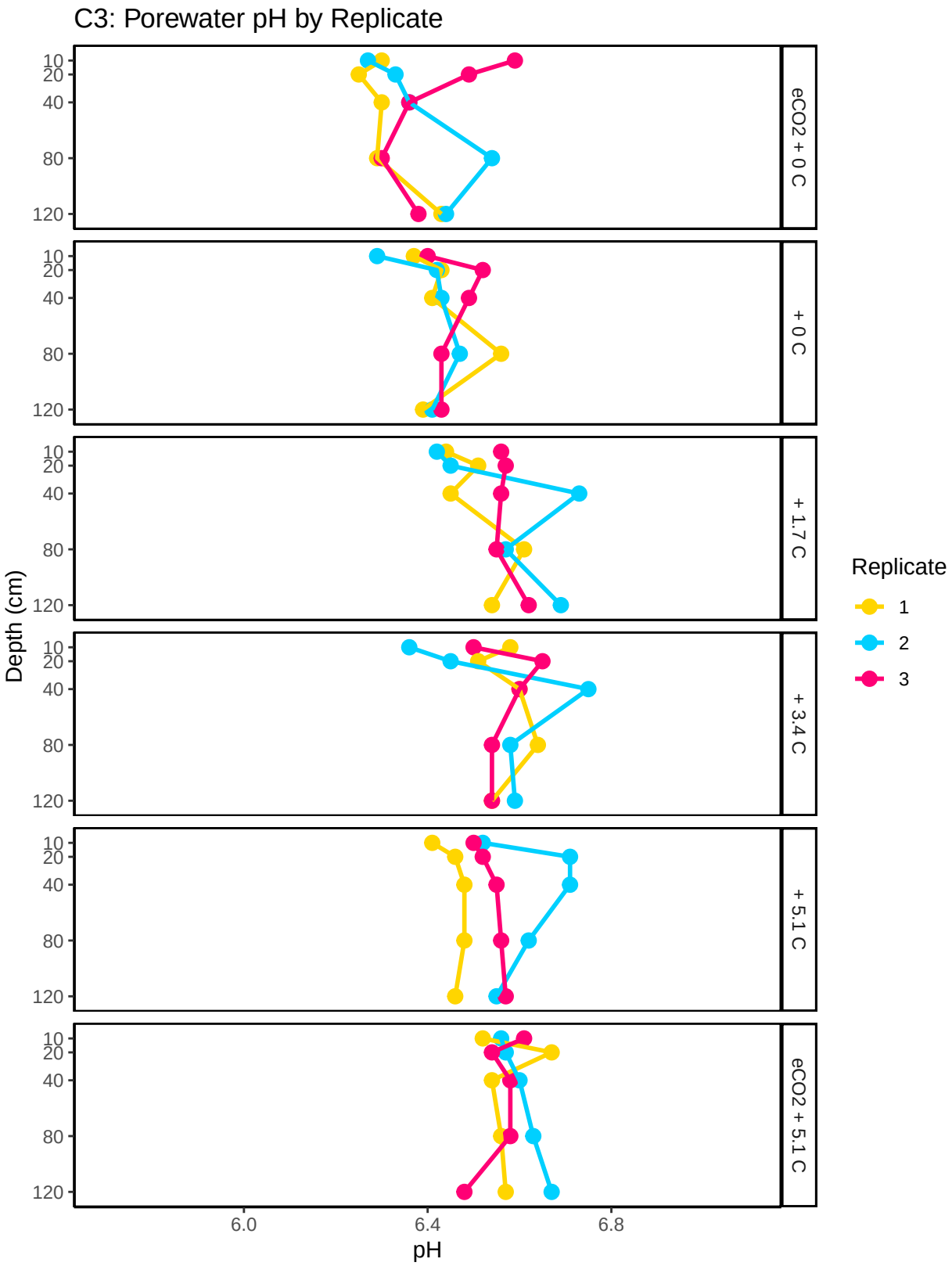
0.2 Visualize Data

```
## Visualize Data
```


C3: Porewater pH
5.1
3.4
1.7
0
10
20
40
80
120
5.5
6.0
6.5
7.0
pH
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2

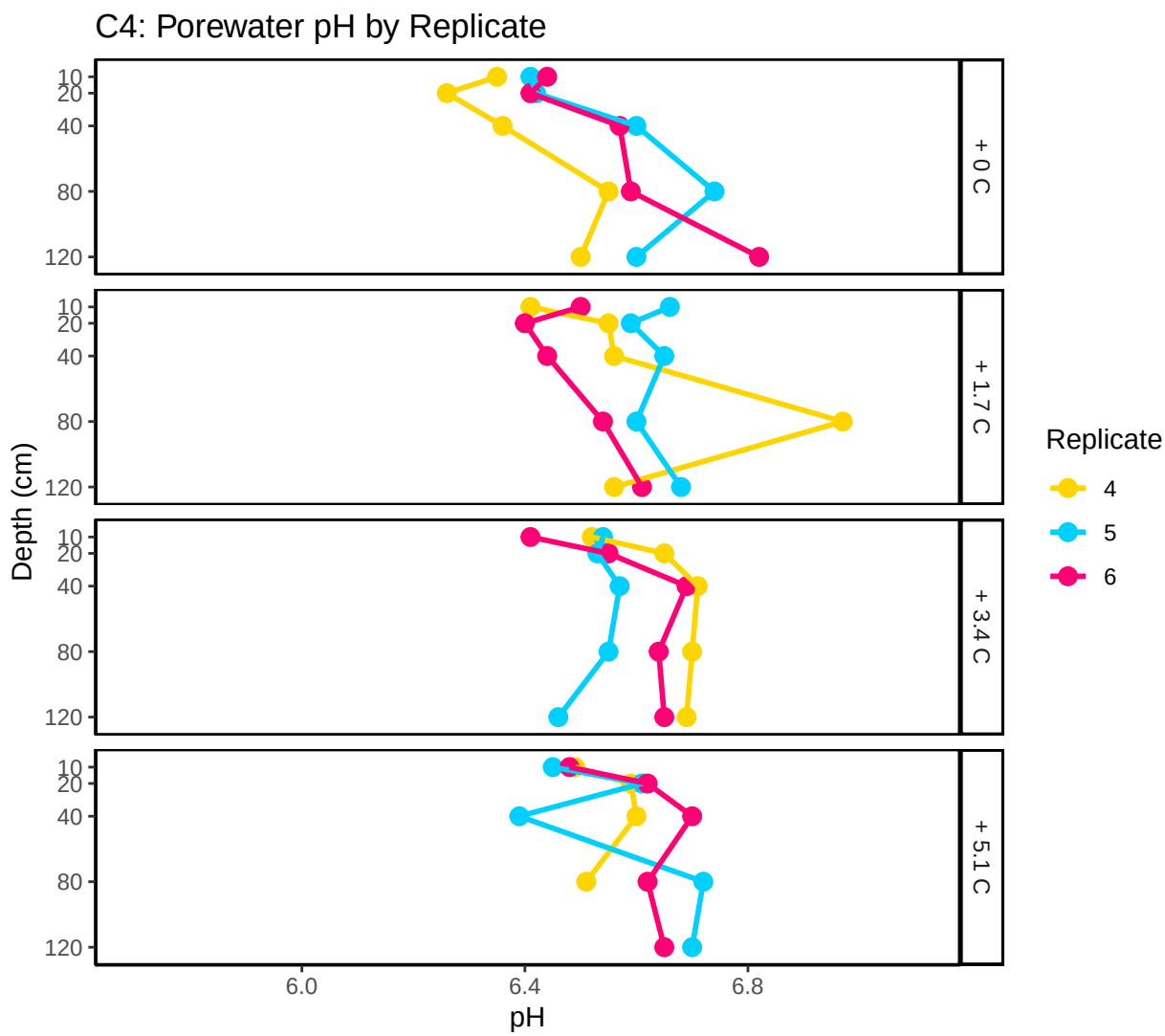

C4: Porewater pH
5.1
3.4
1.7
0
10
20
40
80
120
5.5
6.0
6.5
7.0
pH
+Temp. (C)
0
1.7
3.4
5.1

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```


C3: Porewater pH by Replicate
Depth (cm)
10
20
40
80
120
eCO2 + 0 C
+ 0 C
+ 1.7 C
+ 3.4 C
+ 5.1 C
eCO2 + 5.1 C
6.0
6.4
6.8
pH
Replicate
1
2
3

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_path()`).
```

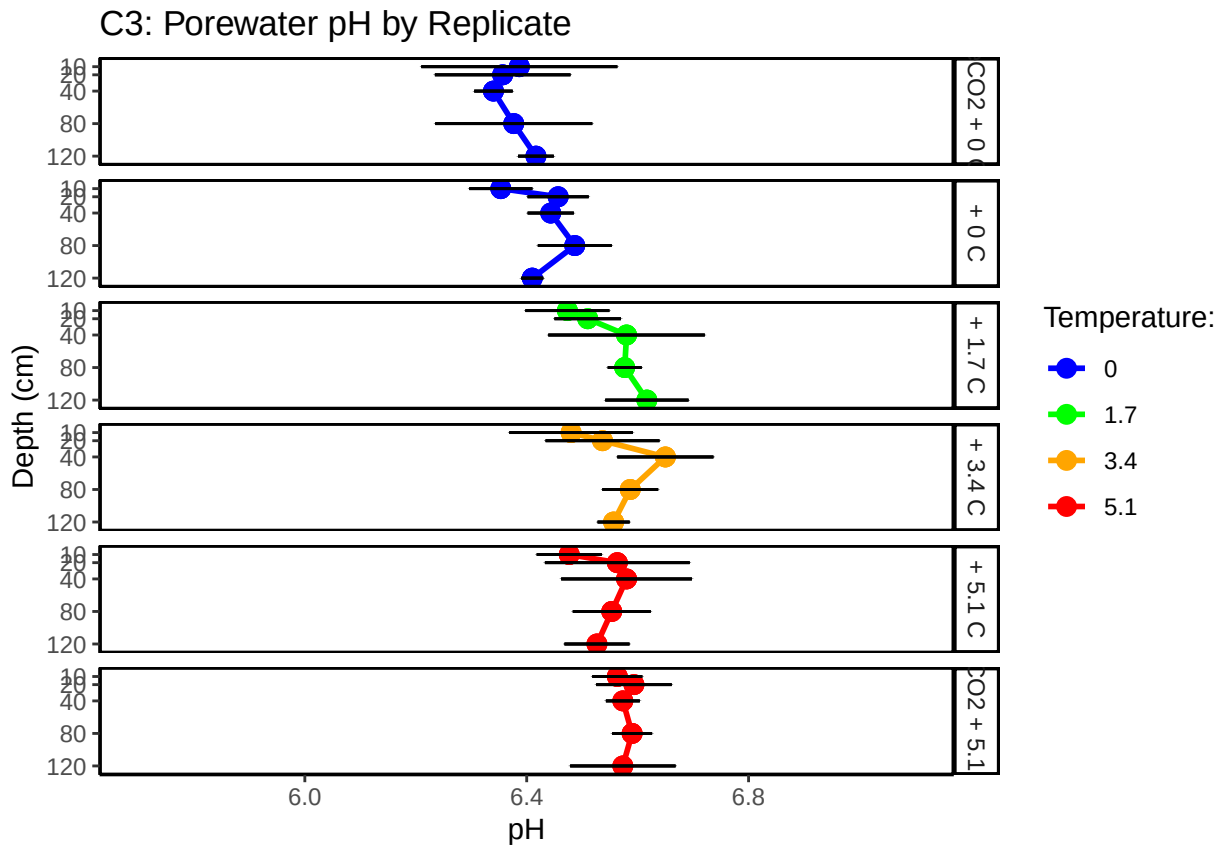



```
pH_forplot_C3_mean <- pH_forplot_C3 %>%
  mutate(Treatment_Depth = paste0(Treatment, "_", Depth_cm)) %>%
  group_by(Treatment_Depth) %>%
  summarize(pH_mean = mean(pH),
            pH_sd = sd(pH),
            across()
            ) %>%
  ungroup
```

```
## Warning: There was 1 warning in `summarize()`.
## i In argument: `across()`.
```

```
## Caused by warning:
## ! Using `across()` without supplying `.cols` was deprecated in dplyr 1.1.0.
## i Please supply `.cols` instead.

## Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
## dplyr 1.1.0.
## i Please use `reframe()` instead.
## i When switching from `summarise()` to `reframe()`, remember that `reframe()`
##   always returns an ungrouped data frame and adjust accordingly.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## `summarise()` has grouped output by 'Treatment_Depth'. You can override using
## the `.groups` argument.
```

```r
#ensure points are in correct order
pH_forplot_C3_mean <- pH_forplot_C3_mean[order(pH_forplot_C3_mean$Depth_cm), ]

#make plot
viz_pH_plot_C3_22 <- ggplot(data = pH_forplot_C3_mean,
                            aes(x = pH_mean, y = Depth_cm, color = Temp_Treatment)) +
  geom_point(size = 3) +
  geom_path(size = 1) +
  geom_errorbar(aes(xmin = pH_mean - pH_sd, xmax = pH_mean + pH_sd),
                width = 0.2, color = "black") +
  scale_color_manual(values = c(
    "0" = "blue",
    "1.7" = "green",
    "3.4" = "orange",
    "5.1" = "red"
  )) +
  facet_grid(factor(
    Treatment1,
    levels = c("eCO2 + 0 C", "+ 0 C", "+ 1.7 C", "+ 3.4 C", "+ 5.1 C", "eCO2 + 5.1 C")
  ) ~ ., scales = "free_x") +
  labs(x = "pH",
       y = "Depth (cm)",
       title = "C3: Porewater pH by Replicate",
       color = "Temperature:") +
  scale_y_reverse(breaks = c(10, 20, 40, 80, 120),
                  limits = c(125, 5)) +
  xlim(5.7, 7.1) +
  theme_classic() +
  theme(
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    panel.border = element_rect(
      color = "black",
      fill = NA,
      linewidth = 0.8
    )
  )

viz_pH_plot_C3_22
```



##Export Data

```
#pull out what we need

df_all_clean_cols <- df_all_clean %>%
  select(
    Site,
    Project,
    Year,
    Month,
    Community,
    Nickname,
    Plot,
    Depth_cm,
    CO2_Treatment,
    Temp_Treatment,
    pH,
    Analysis_rundate
  )

final_data_labeled <- df_all_clean_cols %>%
  mutate(
    Run_notes = run_notes,# new column with notes about the run
    )

#Order Columns for Final Data Frame
All_Clean_Data <- final_data_labeled %>%
```

```r
  select(
    Site,
    Project,
    Year,
    Month,
    Community,
    Nickname,
    Plot,
    Depth_cm,
    CO2_Treatment,
    Temp_Treatment,
    pH,
    Analysis_rundate,
    Run_notes
  )
#head(Samples1)

write.csv(All_Clean_Data, final_path)
```